

Global Considerations in Kidney Disease: Latin America

Jonathan S. Chávez-Íñiguez Sr. | Guillermo García-García | Rolando Claure-Del Granado

CHAPTER OUTLINE

CHRONIC KIDNEY DISEASE IN LATIN AMERICA, 2211

END-STAGE KIDNEY DISEASE IN LATIN AMERICA, 2214

ACUTE KIDNEY INJURY IN LATIN AMERICA, 2217

PEDIATRIC KIDNEY DISEASE IN LATIN AMERICA, 2222

TRENDS AND THE FUTURE, 2222

KEY POINTS

- The lack of national and regional registries limits understanding the true burden of kidney diseases in Latin America (LA).
- LA is one of the regions with the highest prevalence and incidence of kidney diseases, which, can be explained partly by the large number of patients with diabetes, obesity, and hypertension.
- The Mesoamerican region suffers from a particular entity, Mesoamerican nephropathy, which is of unknown etiology, but agrochemicals and high temperatures are the most studied attributable factors.
- The sociocultural and economic conditions of many regions of LA have limited comprehensive care for kidney pathologies; most people have incomplete access to kidney replacement therapies such as dialysis or transplant.
- Acute kidney injury in the Latin American region is predominantly acquired in the community. It has etiologies such as dehydration, heat stroke, insect bites, snakes, and poisoning.
- The epidemiology of subgroups such as pediatrics, geriatrics, and pregnancy is underrepresented in the region.

Latin America (LA) is a geographic region that includes Mexico and the Caribbean Islands in the north to the southern cone shared by Argentina and Chile and covers an area of 20.425 million km.² It includes 20 countries with a growing population of 654 million people and a life expectancy of 72 years.¹ Latin Americans form a diverse ethnic and cultural group of Romance language (Spanish, Portuguese, and French)-speaking people of predominantly European ancestry mixed with indigenous and African roots.²

LA has proved relatively resilient in the face of rising debt stress, inflation, and the COVID-19 pandemic. Income and employment have largely recovered. Current GDP is \$US 5.45 trillion and a gross domestic product (GDP) per capita

of \$US 8327, close to 2019 levels.¹ However, government health care spending in 2019 was estimated at 7.9% of GDP across the region, compared with the Organization for Economic Cooperation and Development health care expenditure of 9.6%. Irrespective of estimate, expenditure varies widely across each country, with Cuba spending 12.4% of its GDP on health care, while Haiti spent just 3.3%³⁻⁵ (Table 74.1). Data on quality of care and health care inequalities remain unavailable for many countries in LA.

In recent decades, the region has experienced a demographic and epidemiologic transition, with reductions in child mortality and communicable diseases, longer life expectancy, and rapid urbanization triggering a surge in lifestyle-associated

Table 74.1 Sociodemographic, Health Expenditure, Nephrologist Manpower, Prevalence, and Access to Kidney Replacement Therapy in Latin America and the Latin Caribbean

Country	Population ^a	GDP Per Capita (US\$)	LEB (years)	Expenditure on Health Per Capita (US\$)	Expenditure on Health (% of GDP)	Nephrologists(pmp)	Access to KRT (%) ^b	Prevalence of Treated ESKD (pmp) ^b
Central America, Mexico, and Latin Caribbean								
Costa Rica	5,123,105	12,670	81	953.08	7.86	5	100	567
Cuba	11,300,698	9126	78	1186.16	12.49	44	100	430
Dominican Republic	10,999,664	8282	73	354.10	4.94	16	67	485
Guatemala	17,362,718	4363	72	289.13	6.47	5	100	575
Haiti	11,306,801	1205	64	45.42	3.31	0.6	N/A	N/A
Honduras	10,121,763	2575	72	212.70	9.04	3	25	405
El Salvador	6,292,731	4187	75	385.74	9.85	9	40	776
Mexico	125,998,302	9946	76	538.57	6.24	9	51	1823
Nicaragua	6,755,895	1913	75	161.20	8.63	5	50	111
Panama	4,294,396	15,728	79	1214.75	9.66	9	80	701
Puerto Rico	3,193,694	32,916	79	N/A	N/A	23	92	2129
Andean Region								
Bolivia	11,936,162	3552	72	241.12	7.86	7	90	457
Colombia	50,930,662	6425	79	477.27	8.99	8	100	858
Ecuador	17,588,595	6223	78	478.53	8.48	12	100	768
Peru	33,304,756	7097	80	388.55	6.3	11	100	618
Venezuela	28,490,453	6613	74	142.46	3.82	18	100	320
Southern Cone and Brazil								
Argentina	45,036,032	9947	77	863.71	9.98	25	100	963
Brazil	213,196,304	8898	76	700.71	10.31	20	100	963
Chile	19,300,315	14,742	81	1278.18	9.75	7	100	1550
Paraguay	6,618,695	5415	76	405.63	7.58	9	100	387
Uruguay	3,429,086	17,768	77	1429.51	9.15	51	100	1194

^aFrom World Health Organization. The Global Health Observatory 2023. <https://www.who.int/data/gho/data/themes/mortality-and-global-health-estimates> and World Health Organization. Global Health Expenditure Database 2023. https://apps.who.int/nha/database/country_profile/Index/en.

^bFrom Luxardo R, Ceretta L, González-Bedat M, et al. The Latin American Dialysis and Renal Transplantation Registry: report 2019. *Clin Kidney J.* 2022;15(3):425–431. ESKD, End-stage kidney disease; GDP, gross domestic product; KRT, kidney replacement therapy; LEB, life expectancy at birth; US\$, U.S. dollars.

Table 74.2 Mortality, DALYs, and YLLs Attributable to CKD in Latin America and Latin Caribbean

Country	Deaths Per 100,000 Population (95% UI)	DALYs Per 100,000 Population (95% UI)	YLL Per 100,000 Population (95% UI)
Central America, Mexico and the Latin Caribbean			
Costa Rica	23.4 (12.4-39.1)	737 (316-1159)	525 (267-899)
Cuba	8.3 (4.5-13.6)	344 (161-527)	209 (108-348)
Dominican Republic	20.8 (9.8-37.9)	671 (211-1130)	523 (238-975)
Guatemala	55.1 (29.9-90.6)	1625 (645-2605)	1403 (733-2361)
Haiti	35.0 (15.4-70.4)	1166 (390-1943)	1000 (425-2026)
Honduras	49.4 (24.6-88.3)	1299 (388-2210)	1115 (534-2029)
El Salvador	72.9 (37.4-124.3)	2122 (693-3551)	1877 (925-3288)
Mexico	36.7 (21.4-55.9)	1210 (656-1763)	911 (514-1415)
Nicaragua	73.9 (40.2-120.2)	2036 (836-3236)	1785 (939-2959)
Panama	23.9 (12.8-39.6)	710 (303-1117)	519 (267-881)
Andean Region			
Bolivia			
Colombia	12.3 (6.5-20.5)	398 (174-621)	251 (129-430)
Ecuador	37.4 (20.4-60.7)	935 (391-1479)	794 (417-1321)
Peru	21.3 (10.7-36.9)	572 (201-943)	464 (225-824)
Venezuela	21.8 (11.5-36.2)	671 (286-1056)	475 (241-811)
Southern Cone and Brazil			
Argentina	21.5 (12.3-33.6)	530 (256-805)	438 (241-700)
Brazil	15.5 (9.3-22.5)	449 (267-631)	336 (199-496)
Chile	13.5 (7.8-21.0)	324 (163-484)	239 (134-379)
Paraguay	27.9 (14.6-46.8)	781 (299-1263)	648 (327-1115)
Uruguay	14.1 (8.8-21.0)	338 (186-490)	270 (166-407)

CKD, Chronic kidney disease; DALY, disability-adjusted life years; UI, uncertainty interval; YLL, years of life lost.

Reproduced from: Pan American Health Organization. The burden of kidney diseases in the Region of the Americas, 2000-2019. Washington, D.C.: PAHO; 2021. Available from: <https://www.paho.org/en/enlace/burden-kidney-diseases> and accessed on 11th May 2025.

noncommunicable diseases (NCD).² Currently, the region has the largest migration crisis in history. In addition to the traditional flows from Central America and Mexico to the United States, Venezuela, and Haiti have been the source of large outflows.¹ This situation is exacerbated by the effects of climate change, which has caused significant economic and social losses. Hurricanes, floods, and droughts are becoming more frequent, and it is estimated that 17 million people could be forced to leave their homes and nearly 5.8 million people could fall into extreme poverty by 2030, mainly due to a lack of safe drinking water, as well as increased exposure to excessive heat and flooding, increasing the risk of acute kidney injury (AKI) and chronic kidney disease (CKD).¹ Additionally, forced migration and climate change represent a serious challenge to host countries to provide renal health care including dialysis and kidney transplantation to an increasingly growing migrant population.^{6,7}

CHRONIC KIDNEY DISEASE IN LATIN AMERICA

Kidney disease has become a public health problem in LA. The prevalence of CKD has been estimated at 10.1%,⁸ with

Haiti reporting the highest (14.0%) and Bolivia the lowest prevalence (6.3%).^{9,10} According to the Global Burden of Disease study (1990–2019), the LA and Caribbean region was estimated to have the greatest burden of CKD among all world regions. Age-standardized disability-adjusted life years (DALYs) rates were highest in Central and Andean LA, at 1348.1 and 836.3 per 100,000 population, respectively, compared with the global rate of 514.9 per 100,000 population.¹¹ CKD was also the leading cause of mortality and disease burden in the region in 2019 (Table 74.2). Kidney disease represented the 8th cause of mortality and the 10th cause of years of life lost (YLLs) and DALYs in both sexes combined, and it was among the causes with the highest increase in the region¹² (Figs. 74.1 to 74.3).

RISK FACTORS FOR CHRONIC KIDNEY DISEASE

The three leading causes of mortality and DALYs in LA, particularly in Central America and the Caribbean, were diabetes, arterial hypertension, and obesity, with an annual increase rate change in all-age DALYs from 2010 to 2019 between 1.6% and 5.0%.¹³

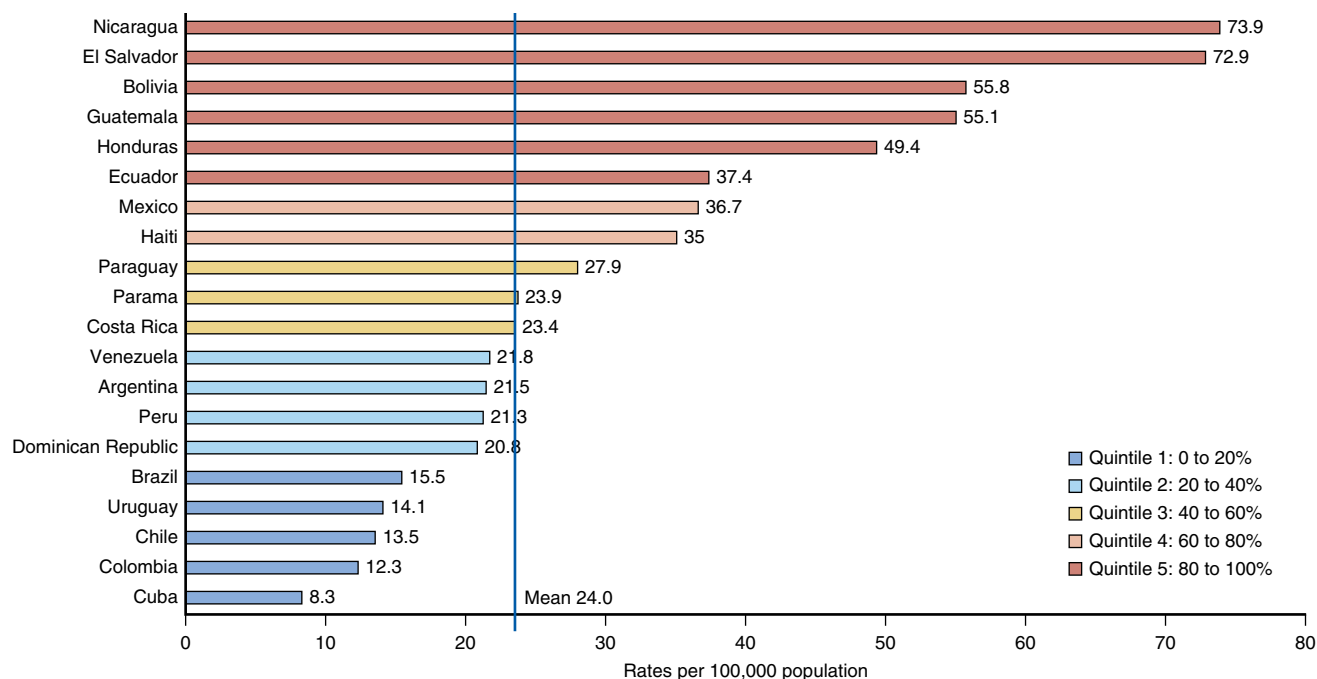


Fig. 74.1 Age-standardized death rates due to CKD in Latin America and the Latin Caribbean, per 100,000 population in 2019. (Reproduced from: Pan American Health Organization. The burden of kidney diseases in the Region of the Americas, 2000-2019. Washington, D.C.: PAHO; 2021. Available from: <https://www.paho.org/en/enlace/burden-kidney-diseases>; and accessed on 11th May 2025.)

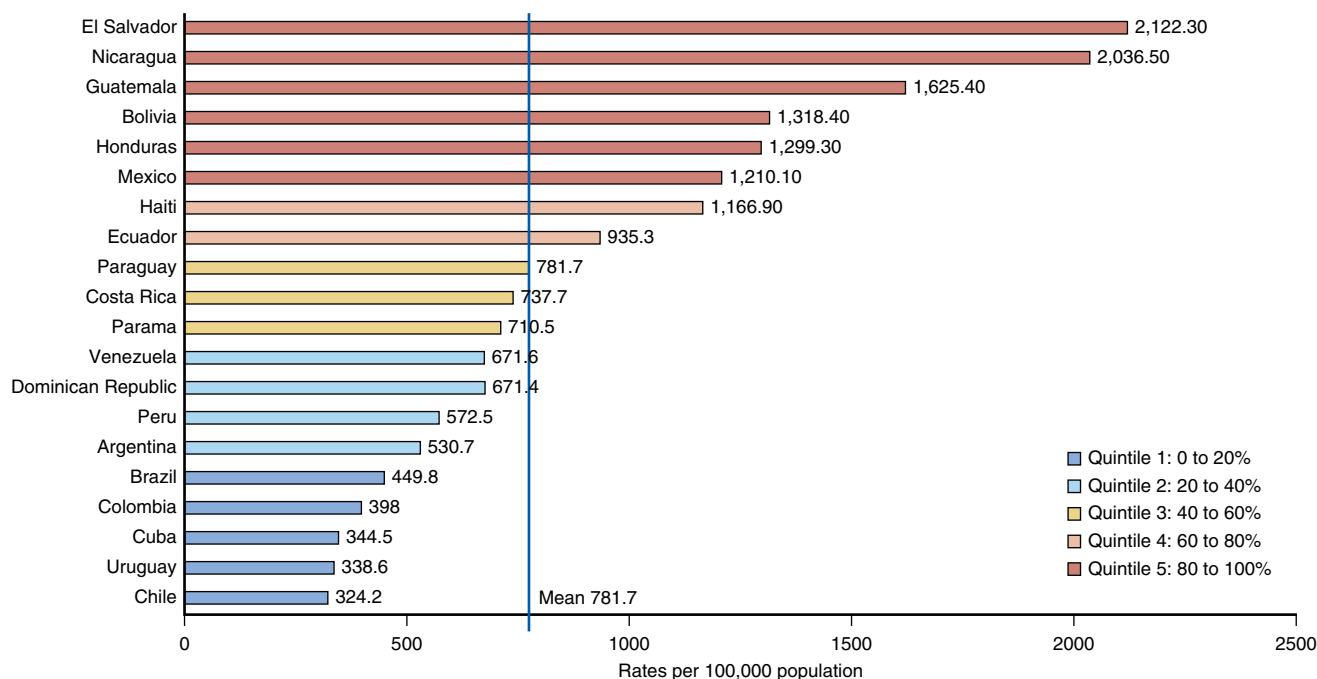


Fig. 74.2 Age-standardized disability-adjusted life years (DALYs) due to chronic kidney disease in Latin America and the Latin Caribbean, per 100,000 population in 2019. (Reproduced from: Pan American Health Organization. The burden of kidney diseases in the Region of the Americas, 2000-2019. Washington, D.C.: PAHO; 2021. Available from: <https://www.paho.org/en/enlace/burden-kidney-diseases> and accessed on 11th May 2025.)

CHRONIC KIDNEY DISEASE RELATED TO DIABETES IN LATIN AMERICA

Diabetes is by far the leading cause (36%) of treated end-stage kidney disease (ESKD), representing 67%, 58%, and 48% of incident dialysis patients in Puerto Rico, Mexico, and Paraguay, respectively, followed by hypertension and

chronic glomerular disease.⁹ On average, the prevalence of diabetes in LA countries was 9.7%. Mexico was the country that experienced the largest increase in diabetes prevalence, of 4.40 percentage points, while the prevalence in Venezuela and Uruguay decreased around 6 percentage points in the 2010–2019 period.¹⁴ The burden of diabetes

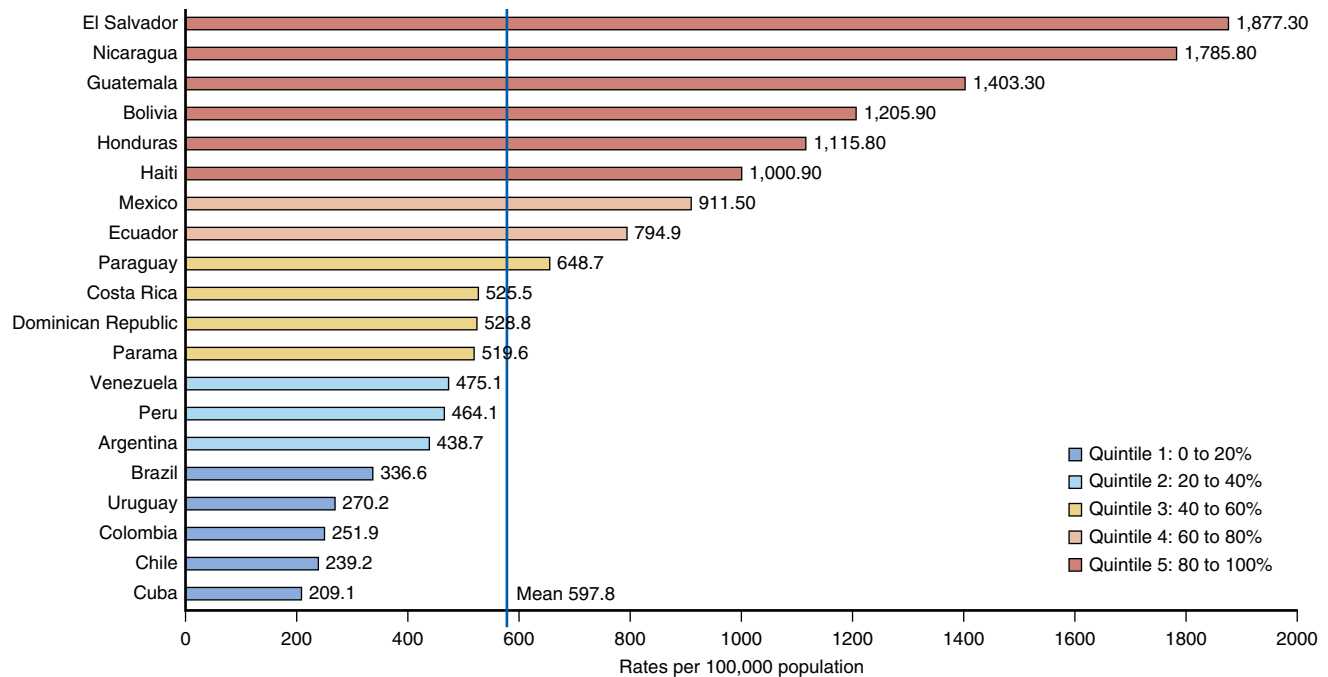


Fig. 74.3 Age-standardized years of life lost (YLLs) due to chronic kidney disease in Latin America and the Latin Caribbean, per 100,000 population in 2019. (Reproduced from: Pan American Health Organization. The burden of kidney diseases in the Region of the Americas, 2000-2019. Washington, D.C.: PAHO; 2021. Available from: <https://www.paho.org/en/enlace/burden-kidney-diseases> and accessed on 11th May 2025.)

and kidney diseases is particularly high compared with the global average, equivalent to 2560 DALYs per 100,000 population compared with a global average of 1460 DALYs per 100,000 population.¹⁵

CHRONIC KIDNEY DISEASE RELATED TO HYPERTENSION IN LATIN AMERICA

Hypertension is the most important risk factor contributing to the burden of cardiovascular disease and the leading cause of death in all LA countries. Close to 22% of the adult LA population is estimated to be affected by hypertension, below the global average of 30%. Peru and Panama have the highest prevalence of 25%, while the lowest prevalence was observed in the Dominican Republic (17.9%) and Paraguay, the only country below 15%. Between 2005 and 2015, most LA countries reduced the prevalence of raised blood pressure with an average of -8%. Changes in risk factors and improvements in detection and treatment of raised blood pressure have, at least partly, contributed to these general reductions.^{14,16}

CHRONIC KIDNEY DISEASE RELATED TO OBESITY IN LATIN AMERICA

Overweight and obesity are major public health concerns in the region. Obesity is an established risk factor for hypertension, diabetes, and cardiovascular disease. In LA countries, 63% of women and 53% of men are overweight (overweight + obesity).¹⁴ Between 2010 and 2016, the overweight population increased 6% in women and 9% in men. In LA countries, obesity is higher among women (29%) than men (18%).¹⁴ Social determinants of health such as

poverty, inadequate water and sanitation, and inequitable access to education and health services underlie the epidemic of obesity and diabetes in the region. A key driver is a changing food environment, in which nutrient-poor and energy-dense processed foods are being aggressively marketed.

MESOAMERICAN NEPHROPATHY—CHRONIC KIDNEY DISEASE OF UNKNOWN CAUSE

An endemic type of CKD of unknown origin (also known as “Mesoamerican Nephropathy” [MeN]) has been described in the region, particularly in El Salvador, Guatemala, Honduras, Nicaragua, Mexico, and Paraguay.^{10,17} This form of CKD has been most consistently reported among agricultural workers,^{18,19} has a higher prevalence in men, favors regions at lower altitudes, and particularly affects workers in sugar cane plantations.¹⁹ This form of CKD of unknown cause, named MeN, is endemic to Pacific coastal communities in Central America.²⁰

The clinical presentation of MeN is characterized by an elevation of serum creatinine level and non-nephrotic proteinuria, predominantly in young men without evidence of any known risk factors for CKD such as hypertension or diabetes, raising the possibility of a tubulointerstitial kidney disease due to an unknown environmental or occupational hazard.²¹⁻²³ Various hypotheses have been described including ischemic injury as a result of the heavy physical work in a hot climate leading to repeated episodes of chronic dehydration with AKI after working hours.²⁴ The temperature in Mesoamerica has increased significantly,²⁵ and recent evidence has shown that kidney tubules can be damaged by heat stress.²⁶ Nonetheless, it is

likely that multiple “hits” are necessary. In kidney biopsies from eight male sugarcane workers (aged 22–57 years) from a rural area of El Salvador who had presented with impaired kidney function (estimated glomerular filtration rate (GFR) between 27 and 77 mL/min/1.73 m²), inactive urinary sediment, and non-nephrotic proteinuria,^{27,28} the morphologic picture was similar in all patients, showing chronic tubulointerstitial injury with tubular atrophy and interstitial fibrosis, and a surprising finding was the presence of relatively extensive glomerular sclerosis.²⁷ These findings suggest primary tubulointerstitial disease.

Other authors have also described acute interstitial nephritis in young people without other comorbidities who had abnormalities in kidney function markers.²⁹

Kidney tubular injury may also be induced by pesticides used in agriculture, which induce proximal tubular injury similar to that observed with calcineurin inhibitors.³⁰ The pesticide paraquat has been shown to be strongly associated with the progression of CKD in exposed individuals in regions of Central America.³¹ Acute heavy metal exposure has also been implicated in Central American communities, especially in a town in Mexico called Poncitlan, where its inhabitants have one of the highest prevalence of CKD of undetermined etiology in the world.³² Another of the most commented hypotheses are those derived from fructose.^{33,34} It is likely that there is also a genetic predisposition for MeN. In a municipality in Mexico where MeN is highly prevalent, genetic variants of ATP6V0A2 were identified in more than 30% of the affected population. ATP6V0A2 encodes the ATPase pump of the lysosomal H⁺ transporter in the kidney tubule.³⁵ Thus it is possible that MeN may be a kidney disease due to a combination of genetic predisposition, with chronic dehydration leading to tubulointerstitial injury^{36,37} and toxins (e.g., pesticides) or heavy metals contributing as second hits in the development of the disease. Although originally reported in Central America, MeN could represent a larger entity, explaining CKD epidemics seen in countries such as Sri Lanka, Bangladesh, and central Australia.³⁸

HEAT STRESS NEPHROPATHY

Heat stress is a syndrome that can occur in hot environments during the summer months.^{39,40} When heat waves persist for consecutive days, the risk of AKI and CKD increases.⁴¹ Dehydration, a common condition among workers in agriculture, construction, firefighters, and miners, is a significant risk factor for nephropathy.³⁹ Recurrent dehydration increases serum osmolality and activates the polyol pathway in the kidney, leading to increased conversion of glucose to sorbitol and fructose.³³ Fructose in turn can be metabolized by the fructokinase that is constitutively present, resulting in the generation of uric acid, oxidants, and chemokines, which can cause local tubulointerstitial injury.³⁴ Mice with knockout of fructokinase are protected from CKD when subjected to repeated dehydration.³³ Heat stress and strenuous physical work can also lead to rhabdomyolysis.⁵³ Continuous episodes of rhabdomyolysis are associated with elevated uric acid levels. This episode of environmentally associated AKI could be related to unknown CKD (CKDu) or MeN as discussed earlier.^{42,43}

END-STAGE KIDNEY DISEASE IN LATIN AMERICA

ESKD continues to grow in the region.^{44–49} In 2019, the median prevalence of treated ESKD in the region was 866 per million population (pmp), compared with 335 pmp in 2000 (Fig. 74.4), and the median number of incident cases was 168 pmp.⁵⁰ Mexico and Puerto Rico have the highest rates of ESKD in the region. In most countries more than 90% of the dialysis patients receive hemodialysis, except for Costa Rica (83.9%) (see Fig. 74.5), Nicaragua (65%), El Salvador (56.1%), where peritoneal dialysis (PD) accounted for more than half of dialysis treatment. Patients living with a functioning kidney graft represented >20% of all treated ESKD patients, but variability is high between countries (Table 74.3). The mean number of nephrologists in the region is 19 pmp but varies widely across LA, from 3.3 pmp in Honduras to 51 pmp in Uruguay. Only Argentina, Cuba, Puerto Rico, and Uruguay have >20 nephrologist pmp¹⁷ (see Table 74.1).

In most countries, a mixture of public and private health care systems provides funding for kidney replacement therapies (KRTs). Currently, 55% of LA countries provide KRT free at the point of delivery (see Table 74.1).^{17,44,45} In the region, the median annual costs per person for maintenance hemodialysis (HD) (\$US 17,454) and maintenance PD (\$US 16,825) were below global averages (\$US 22,617 and \$US 20,524, respectively).^{17,44,45} The annual cost of kidney transplantation in the first year including surgery ranged from \$7458 in Brazil to \$21,207 in Argentina. Except for those countries with universal access to KRT, where patients incurred no costs, out-of-pocket costs for KRT range between 1% and 25%, being higher for transplantation.⁴⁵ The median number of HD centers was 4.55 pmp, with the highest rates reported in Chile, Argentina, and Uruguay (13.0, 13.0, and 11.0 pmp, respectively) and the lowest in Haiti and Guatemala (0.65 and 0.90 pmp, respectively).^{45,46} Home HD is not practiced in the region. In two-thirds of countries, more than half of HD patients began treatment with a temporary dialysis catheter. The median number of PD centers in the region was 0.96 pmp, which was below the global average. Twelve countries (67%) were able to offer adequate frequency of exchanges, whereas just over half had the capacity to measure PD adequacy.^{44,45}

TRANSPLANTATION

Kidney transplantation in LA started in Argentina in 1957, followed by Mexico (1963), Brazil (1964), Chile, and Colombia (1966).⁴² Kidney transplantation is available in 17 (94%) countries, with a regional median of 0.5 transplant centers pmp, above the global median of 0.42 pmp. Mexico and Argentina have the highest capacity for transplantation (2.16 and 1.39 transplant centers pmp, respectively^{44,48}). All countries with transplant capacity perform a combination of deceased and living-donor kidney transplants, except for El Salvador, which has capacity for living-donor kidney transplants only. The kidney transplant rate has doubled over the past 2 decades, from 11 transplants pmp in 2000 to 22 transplants pmp in 2019 (Fig. 74.6). The country-specific

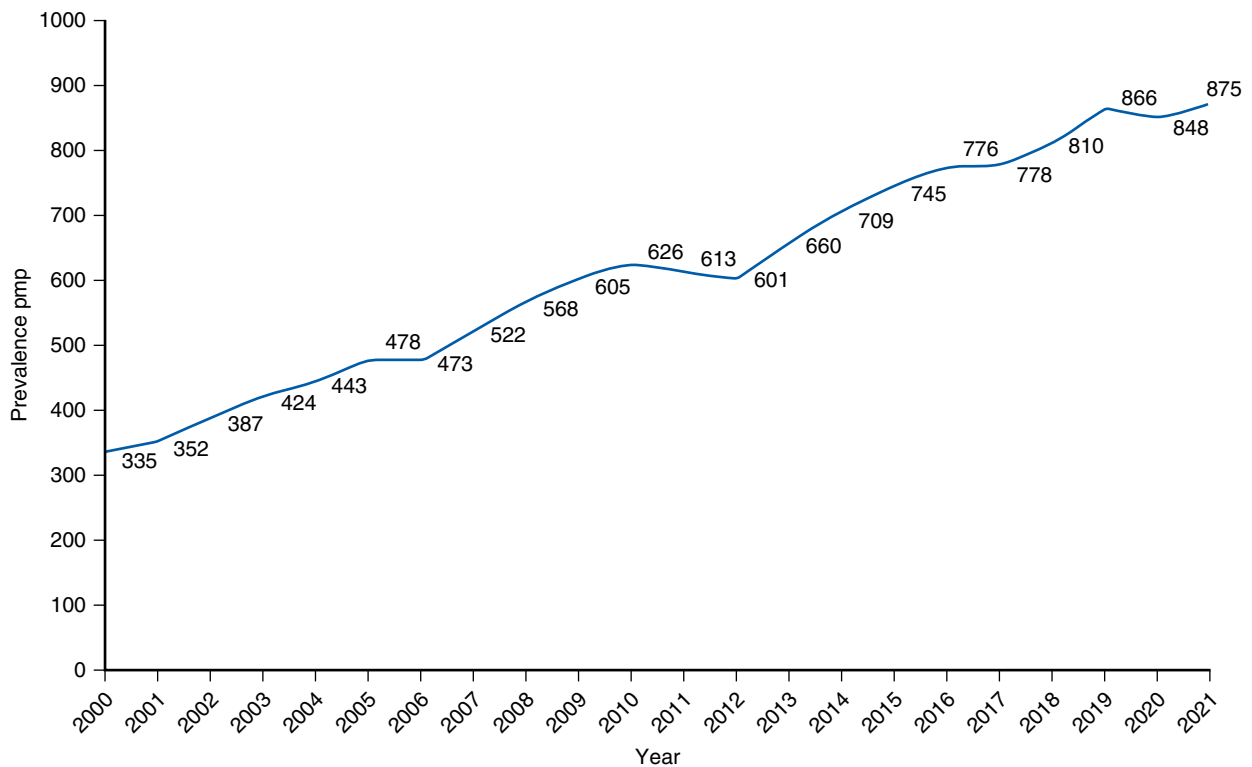


Fig. 74.4 Prevalence of treated end-stage kidney disease and kidney transplantation rates in Latin America and the Latin Caribbean, 2000–2021.

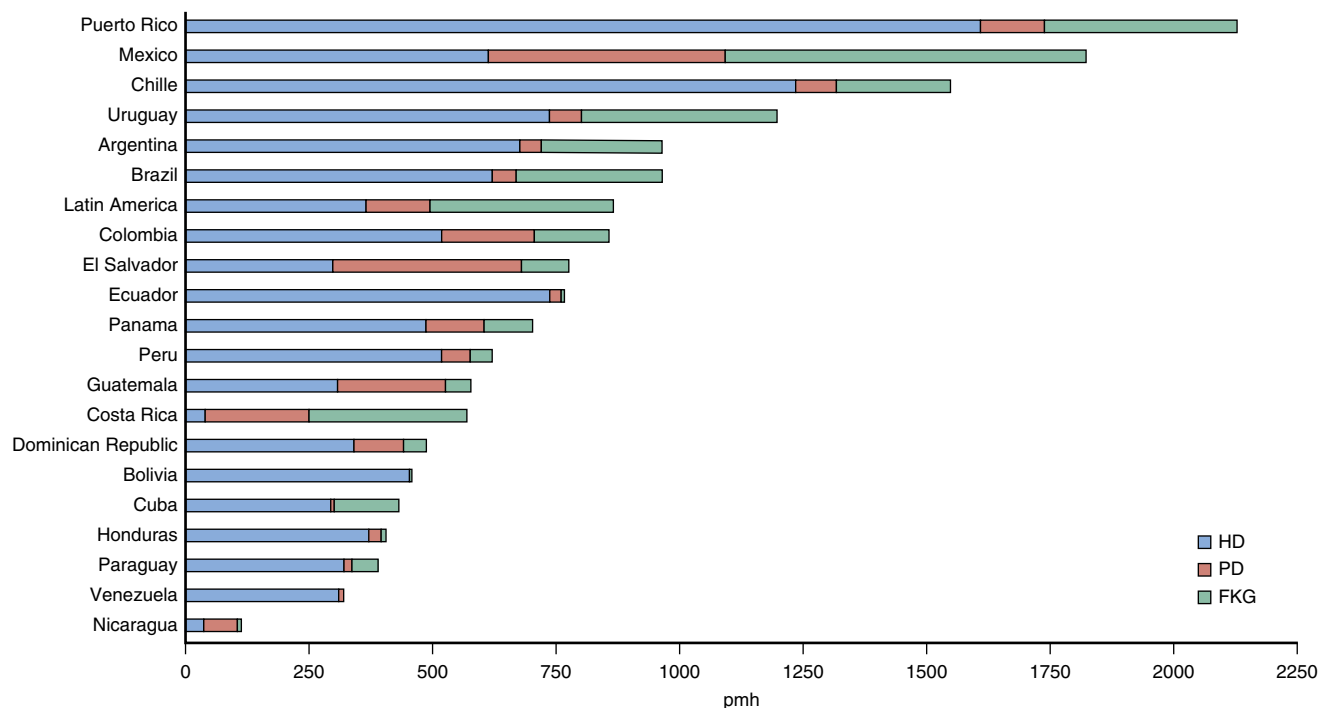


Fig. 74.5 Prevalence of treated end-stage kidney disease by treatment modality in Latin America and the Latin Caribbean, 2019. (From SLANH. Registro Latinoamericano de Diálisis y Trasplante. Reporte 2021. <https://slanh.net/registros/>; 2023. Accessed March 2023.)

Table 74.3 Incidence and Prevalence of Treated End-Stage Kidney Disease by Treatment Modality in Latin America and Latin Caribbean, 2019

					Prevalence Rate (pmp)				
Country	Population	Incidence Rate (pmp)	HD	PD	Total Dialysis	% on PD	FKG	Total Prevalence (pmp)	Kidney Transplant Rate (pmp)
Central America, Mexico, and Latin Caribbean									
Costa Rica	5,123,105	38	40	209	249	83.9	318	567	15
Cuba	11,300,698	108	293	6	299	2.0	131	430	15
Dominican Republic	10,999,664	221	340	98	438	22.3	47	485	5
Guatemala	17,362,718	140	304	221	525	42.0	51	575	6
Haiti	11,306,801	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Honduras	10,121,763	96	370	22	392	5.6	13	405	0
El Salvador	6,292,731	217	297	380	677	56.1	99	776	6
Mexico	125,998,302	530	611	483	1094	44.1	729	1823	62
Nicaragua	6,755,895	31	35	65	100	65.0	11	111	2
Panama	4,294,396	181	488	113	601	18.8	100	701	8
Puerto Rico	3,193,694	419	1607	130	1737	7.4	392	2129	18
Andean Region									
Bolivia	11,936,162	114	452	2	454	0.44	3	457	2
Colombia	50,930,662	103	516	185	701	26.3	157	858	19
Ecuador	17,588,595	6	735	21	756	2.7	12	768	13
Peru	33,304,756	62	515	57	572	9.9	46	618	3
Venezuela	28,490,453	96	310	10	320	3.1	0	320	1
Southern Cone and Brazil									
Argentina	45,036,032	163	674	46	720	6.3	243	963	35
Brazil	213,196,304	218	618	47	665	7.0	299	963	30
Chile	19,300,315	204	1236	81	1317	6.1	233	1550	22
Paraguay	6,618,695	36	317	16	333	4.8	54	387	4
Uruguay	3,429,086	185	734	62	796	7.7	398	1194	42

FKG, Functioning kidney graft; HD, hemodialysis.

Modified from Luxardo R, Ceretta L, González-Bedat M, et al. The Latin American Dialysis and Renal Transplantation Registry: report 2019. *Clin Kidney J*. 2022;15(3):425–431.

deceased organ transplantation rate is heterogenous, with Uruguay, Brazil, and Argentina reporting the higher rates of deceased-donor kidney transplantation in the region (33.8, 20.8, and 16.0 pmp, respectively), and >70% of their transplants were from deceased donors. Most countries (63%) with kidney transplantation available had national transplant waitlists, whereas the rest had regional lists only.^{17,44,47}

The first legislations on organ procurement and transplantation were issued by Brazil (1968) and Uruguay (1971). Currently, most countries have transplantation legislation, and all of them abide to the Declaration of Istanbul to prevent organ trafficking and commercialism. Unrelated living-donor transplantation is forbidden in Argentina and Uruguay but is allowed with legal restrictions in Brazil and Mexico. Transplant for patients from foreign countries is prohibited in Uruguay and Brazil and is practiced with restrictions in Mexico, Chile, Ecuador, and Colombia.⁴⁷

As expected, the COVID-19 pandemic was associated with an important reduction in the incidence and prevalence

of KRT in the region, mostly at the expense of kidney transplantation activity. Incidence and prevalence of KRT rates dropped from 168 pmp and 866 pmp in 2019, to 158 pmp and 848 pmp, respectively, in 2020.⁴⁵ Nevertheless, by 2021 KRT rates had largely recovered and were above the 2019 rates (179 pmp and 875 pmp, respectively).⁴⁵

DIALYSIS AND TRANSPLANTATION REGISTRIES

Fourteen countries (78%) have official dialysis registries, and 13 countries (72%) have transplant registries. Only 6 countries have CKD registries and detection and prevention programs (Argentina, Bolivia, Chile, Colombia, Dominican Republic, Ecuador, and Uruguay).^{10,44} Only Argentina, Chile, Colombia, and the Dominican Republic have national strategies to improve CKD detection and prevention, either as a part of noncommunicable disease programs or CKD-specific policies.^{44,45} Several advocacy groups exist across the region, mainly for the support of patients with CKD. Frequent barriers and limitations for optimal care of patients with kidney disease include lack of

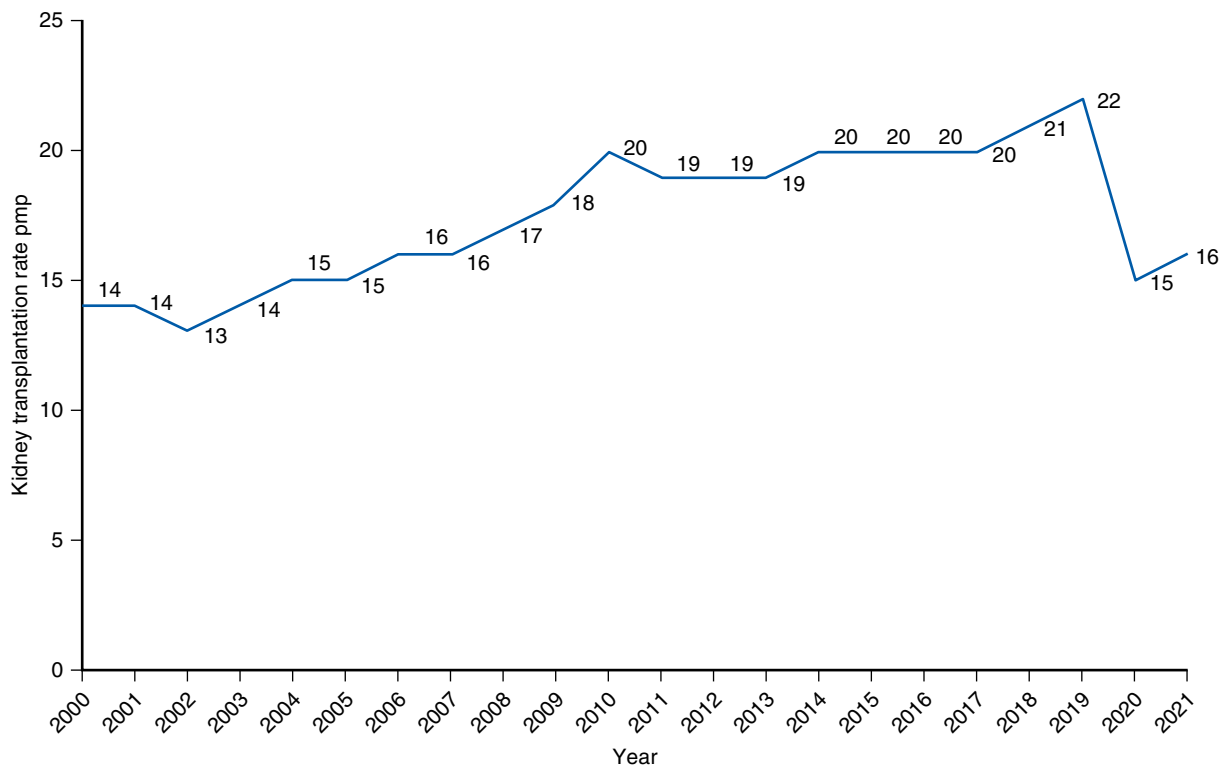


Fig. 74.6 Kidney transplantation rates in Latin America and the Latin Caribbean (2000–2021).

active involvement of patients, inadequate technologic and human resources, geographic imbalance of nephrologists' distribution, improper implementation of clinical guidelines at the operational level, and the fragmentation of the health system.^{10,44,45}

ACUTE KIDNEY INJURY IN LATIN AMERICA

In LA, AKI constitutes a significant health care concern, with substantial impacts on public health and health care systems.⁴⁶ LA faces unique challenges related to AKI due to limited access to medical care and varying socioeconomic conditions.

In LA the climate is characterized mostly by high temperatures, which promotes the survival of pathogenic microorganisms, parasites, and vermin.⁴⁷ Water scarcity in arid regions increases the susceptibility of kidneys to even relatively minor insults. Additionally, the tropics are home to many poisonous snakes and arthropods.^{48,49} Furthermore, due to overcrowding and inadequate living conditions in the tropics, transmission through direct contact and aerosols is more likely.^{49,50} Underreporting of AKI events is likely, especially for community-acquired AKI (CA-AKI). Addressing AKI in LA requires coordinated efforts from governments, health care organizations, and the medical community. This includes improving access to medical care, promoting awareness of risk factors and preventive measures, as well as investing in medical education and health care infrastructure.

AKI in the tropics is usually a single disease, following a specific event (e.g., snakebite) or a specific infection (e.g., malaria), and not a part of a general multiorgan failure

syndrome.⁵⁰ The causes of CA-AKI, which is the most common form of AKI in the LA region can be broadly divided into those caused by infections; animal, plant, or chemical toxins; environmental factors; and obstetric complications (Table 74.4).⁵⁰ Diarrheal illness and malarial AKI are major public health threats in most tropical countries, while leptospirosis, dengue fever, envenomation, and obstetric complications are the main causes of AKI in LA countries.⁵⁰ AKI in the tropics can be part of a constellation of manifestations, presenting as specific patterns of AKI "syndromes," which help the clinician in developing a diagnostic and therapeutic approach (Fig. 74.7).

The scarcity of reports on AKI from LA has impeded efforts to assess its prevalence and compare its patterns, presentation, and outcomes with other regions worldwide.⁴⁹ In an effort to better understand the epidemiology of AKI in the region, the Latin America Society of Nephrology and Hypertension (SLANH) AKI committee has conducted an observational, prospective, longitudinal, multinational cohort study addressed to determine outcomes of AKI in the region.⁵⁰ Across 15 countries, 905 critically ill patients (61% male) were included from tertiary referral centers. Sixty-two percent had CA-AKI, 77% of patients were assessed by nephrologists, KRT was performed in 29%, and in-hospital mortality was 26.5%.⁵⁰

ACUTE KIDNEY INJURY IN OBSTETRICS

The incidence of AKI in pregnancy (PRAKI) in LA is estimated between 4% and 26%.^{51,52} Among critically ill obstetric patients, PRAKI is the most frequent complication, observed in 24% to 28% of women with a maternal mortality of 8.7% to 20.6%.^{53,54} Dialysis requirement for PRAKI is in intensive care

Table 74.4 Causes of Acute Kidney Injury in the Tropics

Vector-borne infections	Malaria (<i>Plasmodium falciparum</i> , <i>Plasmodium vivax</i> , or <i>Plasmodium knowlesi</i> , transmitted by <i>Anopheles</i> mosquito) Dengue fever (dengue virus, transmitted by <i>Aedes</i> mosquito) Scrub typhus (<i>Orientia tsutsugamushi</i> , transmitted by trombiculid mites) Hemorrhagic Rift Valley fever (RVF virus, transmitted by <i>Aedes</i> or <i>Culex</i> mosquitoes)
Direct infections	Leptospirosis (<i>Leptospira interrogans</i>) Hantaviruses, also known as hemorrhagic fever with renal syndrome (Puumala and Hantaan viruses)
Diarrheal diseases	Zygomycosis Viral diarrhea (rotavirus, Norwalk agent) Enterotoxigenic or enteroinvasive <i>Escherichia coli</i> Bacillary (<i>Shigella</i>) dysentery Cholera
Other infections	Melioidosis (<i>Burkholderia pseudomallei</i>) Typhoid (<i>Salmonella typhi</i>) Chlamydia (<i>Chlamydia trachomatis</i>) Legionellosis (<i>Legionella pneumophila</i>) Human immunodeficiency virus
Plant and fungal toxins	Herbal medicines Impila tuber Djenkol beans Marking nut Mushroom Starfruit and other oxalate-rich plants Plant-derived toxins used as insecticides and to kill fish
Animal poisons	Snakebites Wasp, hornet, and bee stings Spider bite Jellyfish sting Scorpion sting Carp gallbladder or bile
Chemical nephrotoxins	Ethylene glycol, propylene glycol Paraphenylenediamine Ethylene dibromide Copper sulfate Chromic acid
Environmental factors	Heat stroke
Obstetric complications	Natural disasters Postabortal sepsis Antepartum and postpartum hemorrhage Preeclampsia

cohorts, ranging from 7.6% to 62%. Kidney recovery varied widely from 39% to 90%, with dialysis dependence ranging from 1.2% to 54%.^{53,54} In a Mexican cohort of PRAKI, 10 of 27 patients had preexisting CKD; 18 patients required dialysis, and most had preterm, small, or very small babies.⁵² These conditions are estimated to be similar across LA.

INFECTIONS

The interplay between infections and AKI poses a considerable burden on the health care system and public

health initiatives throughout the region. Several infectious diseases prevalent in LA can lead to AKI. Among them, two notable examples are Dengue fever and leptospirosis.⁵⁵

DENGUE FEVER

Dengue fever (DF) is a mosquito-borne infection that causes a severe flulike illness and, sometimes, a potentially lethal complication called dengue hemorrhagic fever (DHF). DF and DHF are primarily diseases of tropical and subtropical areas and therefore prevalent in LA.⁵⁶ Brazil and Mexico have the highest incidence with >100,000 cases between 2010 and 2016.^{57,58} The high incidence of DF in LA has been ascribed to global warming and ecosystem modifications, facilitating the dissemination of pathogens and vectors of the disease.⁵⁹ In addition, demographic factors including uncontrolled migration of people and unplanned urbanization have facilitated the rise of DF in urban areas and large cities.

Dengue virus infection may manifest clinically as an undifferentiated fever—DF, DHF, or dengue shock syndrome (DSS). DHF is a severe form of the disease and manifests with fever, hemorrhagic phenomena, thrombocytopenia, and evidence of plasma leakage (high hematocrit, pleural effusion, ascites, and hypoalbuminemia), occurring in patients who have previously been infected with other serotypes of dengue virus (“secondary infection”).⁶⁰ Classic DF is an acute febrile illness with two of the following criteria in a patient coming from an endemic area: nausea and vomiting, a maculopapular rash, aches and pains, positive tourniquet test, and leucopenia.⁶¹ Diagnosis can be made with serology or reverse transcription polymerase chain reaction (RT-PCR).

Kidney Involvement in Dengue Fever

Kidney manifestations of DF include AKI, glomerulonephritis, or hemolytic-uremic syndrome (HUS). The clinical presentation is variable, ranging from AKI, proteinuria, and active urinary sediment to thrombotic microangiopathy.

DF can also cause functional alterations of the renal tubule, with the appearance of hypokalemia and hyponatremia.⁶² The possible pathogenesis of renal involvement includes ischemic or hemoglobin-associated tubular injury, glomerular injury, secondary immune complex deposition, and thrombotic microangiopathy.⁶³

The prevalence of AKI in dengue infection is variable, ranging from 0.3% to as high as 35%,^{64,65} and is considered one of the most common causes of AKI community-acquired in the tropics in LA.⁴⁷ It was reported in 9 cohorts that the presence of AKI was associated with a significant increase in the probability of dying; close to 50% die.⁶⁴ Unfortunately, complete kidney recovery is not achieved in all patients, and it has been observed that 22% of them develop CKD in a 3-month follow-up.^{64,65}

In patients with CKD, the dengue syndrome is more severe and lethal;⁶⁶ similar to patients with kidney transplants, they present more severity of symptoms and a greater probability of dying.⁶⁷

There is no specific treatment for DF, so management consists of supportive care in many places of LA including controlling fever (acetaminophen), preventing dehydration (oral or intravenous fluids), and managing complications such as bleeding. Aspirin and nonsteroidal anti-inflammatory drugs should be avoided. No vaccines

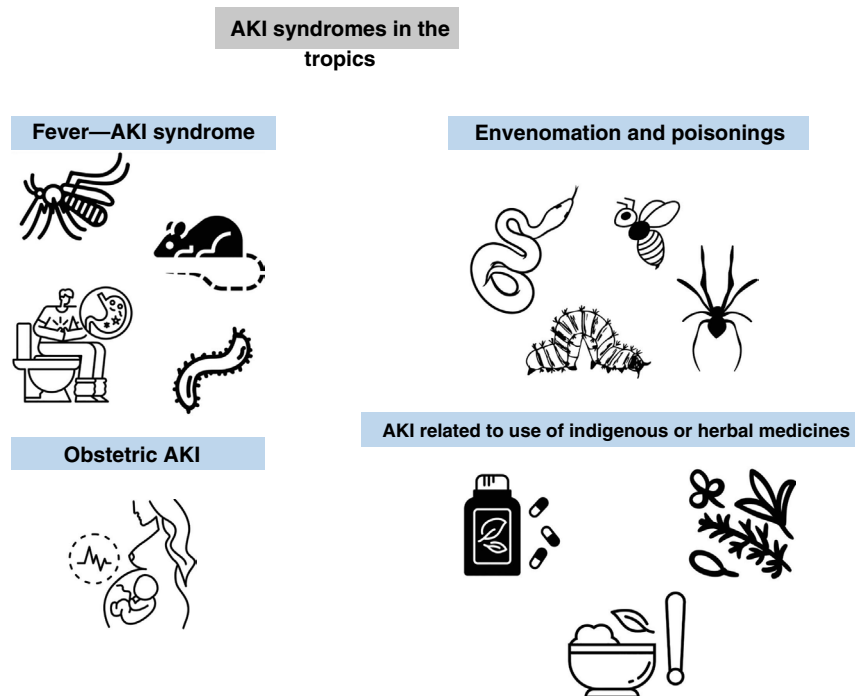


Fig. 74.7 Community-acquired acute kidney injury (AKI) in the tropics can be part of a constellation of manifestations, presenting as specific patterns of AKI syndromes, which help the clinician in deciding a diagnostic approach.

are available. The principal method for prevention of DF is mosquito control.

LEPTOSPIROSIS

Leptospirosis, caused by the spirochete *Leptospira interrogans*, is one of the most important zoonotic bacterial infections worldwide. The global incidence of leptospirosis by 2015 amounted to 1.03 million cases annually with 58,900 deaths, and most cases occur in the tropics.⁶⁸ It is endemic in LA in humid tropical and subtropical areas and usually reaches epidemic levels after higher rainfall periods with flooding or natural disasters such as hurricanes. Humans are only incidentally infected, through abrasions on the skin and mucous membranes and conjunctiva coming into contact with water contaminated with infected rat urine in a *Leptospira*-contaminated environment caused by, for example, local agricultural practices, as well as poor housing and waste disposal, all of which give rise to many sources of infection.^{69,70} Some studies have reported a seroprevalence of anti-*Leptospira* antibodies in up to 33% in asymptomatic individuals.⁷¹ Leptospirosis can provoke a broad range of manifestations, from benign infection, characterized by nonspecific symptoms including abrupt onset of fever, rigors, myalgias, and headache in 75% to 100% of patients after an incubation period of 2 to 26 days (average, 10 days). After the acute septicemic stage, hyperbilirubinemia occurs in 85% of cases.⁷² It is estimated that only 10% developed a severe form of leptospirosis that causes jaundice, hemorrhagic events, and AKI, called Weil syndrome,^{73,74} which carries a mortality rate of 5% to 15%.⁷⁵ On physical examination, conjunctival suffusion is an important sign. Most patients have muscle tenderness, splenomegaly, lymphadenopathy, pharyngitis, hepatomegaly, muscle rigidity, abnormal respiratory auscultation, and/or rash. In patients with

suspected infection, a simple three-point clinical score—the SPiRO score—appears to reliably identify patients at risk of severe disease. An absence of hypotension, oliguria, or abnormal auscultatory findings—a SPiRO score of zero—identifies patients at low risk of severe disease.⁷⁶ The most widely used method for diagnosing leptospirosis is the microscopic agglutination test (MAT), which is performed using two blood samples collected 2 weeks apart or RT-PCR.

Kidney Involvement in Leptospirosis

Nearly all mammals can serve as carriers of leptospirosis, harboring and shedding the organisms from the proximal tubules of the kidneys.⁷⁷ Clinical manifestations vary from urinary sediment changes and electrolyte abnormalities such as hypokalemia to AKI, which is observed in 44% to 67% of patients.⁷⁸ Primary injury of the proximal convoluted tubules⁷⁹ affects ion transport in proximal tubules by inhibition of the Na/K-ATPase pump. Patients may present with high urinary fractional excretion of sodium and potassium.⁷⁸ Pathologic findings in leptospirosis are characterized by acute interstitial nephritis and acute tubular necrosis.⁷⁹ Vasculitis is also observed in the acute phase of the disease.⁷⁹

The vast majority of infections with *Leptospira* are self-limiting. Antibiotic treatment is considered efficient in all stages of the disease. Mild cases can be treated with oral amoxicillin (30–40 mg/kg/day) or ampicillin (50–100 mg/kg/day) four times a day for 7–10 days. Children >8 years of age can receive doxycycline (2 mg/kg/dose) two times a day for 7–10 days. Patients with severe infections should be treated with intravenous penicillin in doses of 50,000–100,000 U/kg/day for 7–10 days or third-generation cephalosporins or erythromycin.⁸⁰ Doxycycline prophylaxis appears to be protective, reducing morbidity

and mortality during outbreaks. Control of reservoirs is difficult, and no vaccines are available. Surface-exposed proteins and putative virulence determinants have been identified that may serve as subunit vaccine candidates. The treatment of Weil syndrome consists of life support measures and antibiotics.⁸¹ Critical ill patients may require KRT and ICU monitoring. The modality of KRT may not affect clinical outcomes in severe leptospirosis.^{82,83} Conservative fluid intake and approaches to minimize lung injury and volume overload. In Brazil, the death rate has declined from 334 in 2015 to 234 in 2016 and to 231 in 2017.⁸⁴

YELLOW FEVER

Yellow fever, caused by the yellow fever virus, is endemic to Africa and South America. The vectors are blood-feeding mosquitoes, primarily belonging to the *Haemagogus* genus in South America.⁸⁵ The spectrum of yellow fever encompasses a range of manifestations, varying from an asymptomatic or mild affliction to a rapidly advancing, life-threatening multiorgan ailment. A substantial proportion of cases exhibit a self-limited ailment, characterized by fever, bradycardia, chills, diminished appetite, headaches, as well as muscular and back discomfort, often leading to a state of prostration.⁸⁶ Around 25% of patients progress to the most severe manifestation of the disease, marked by hepatic failure, jaundice, tendencies toward hemorrhage, epigastric pain, vomiting, shock, and AKI,⁸⁶ with a mortality as high as 50%.⁸⁷ Knowledge about the prevalence, contributing factors, and underlying mechanisms of yellow fever-associated AKI remains limited.⁸⁸ Oliguria often appears after 5 to 7 days.⁸⁹ Kidney biopsies revealed acute tubular necrosis. The precise mechanisms of AKI remain enigmatic. Possible explanations encompass renal ischemia, intravascular coagulation, shock-induced impairment, bilirubin-mediated tubular toxicity, and an extensive release of inflammatory cytokines.⁸⁹

OTHER INFECTIOUS CAUSES OF ACUTE KIDNEY INJURY

Malaria, hantavirus, Chagas disease, and bacterial and viral respiratory infections also have the potential to cause AKI. Malaria, transmitted by *Anopheles* mosquitoes, is endemic in some regions of LA (Table 74.4).⁹⁰ *Plasmodium falciparum* causes the largest burden of disease globally, followed by *Plasmodium vivax*, which is more common in the Americas. Severe cases of malaria can result in AKI due to the release of toxins from damaged red blood cells and the deposition of malaria pigment in the kidney tubules, impairing kidney function.^{90,91}

Hantavirus infections are typically transmitted through contact with infected rodents or their excretions.⁹² In severe cases, hantavirus can lead to hantavirus pulmonary syndrome, characterized by respiratory distress and AKI, as a consequence of endothelial cell damage and increased vascular permeability. Chagas disease, caused by the *Trypanosoma cruzi* parasite, is transmitted mainly through triatomine bugs.⁹³ The chronic form of Chagas disease can lead to cardiac and gastrointestinal involvement, but it can also cause AKI due to the parasite's direct invasion of kidney tissues.⁹³ In the context of bacterial and viral respiratory infections, severe cases of pneumonia and acute respiratory

distress syndrome (ARDS) can lead to sepsis, a major risk factor for AKI. Infections like COVID-19, caused by the SARS-CoV-2 virus, have also been associated with a higher incidence of AKI in LA, particularly among vulnerable populations.⁹⁴

Preventive strategies play a critical role in mitigating the burden of infections and AKI in LA. Vector control programs to manage mosquito and rodent populations, public health campaigns promoting personal hygiene and sanitation, and initiatives to improve water quality are vital. Furthermore, raising public awareness and educating health care professionals about the early recognition of infectious diseases and AKI are crucial. Infections and AKI are intertwined public health challenges in LA.

ENVENOMATION FROM ANIMALS AND INSECTS

Encounters with animals and insects are frequent in tropical regions due to their widespread presence, the occupational activities primarily focused on farming and the insufficient availability of protective gear. Envenomation resulting from snakebites and insect stings contributes significantly to the incidence of AKI in LA. While the bite or sting is often apparent, it is crucial to consider this diagnosis even in the absence of an overt history, particularly when clinical manifestations suggest it. This consideration is equally important when dealing with cases of AKI in children.

NEPHROTOXICITY DUE TO VENOMS

Spiders (*Loxoscelism*)

Spiders of the genus *Loxosceles* are known colloquially as recluse spiders, violin spiders, fiddleback spiders, and, in South America, by the nonspecific name “brown spiders.”⁹⁵ *Loxosceles* spiders are of medical importance in Brazil and Chile. The most common species involved in envenomations are *Loxosceles laeta*, *Loxosceles intermedia*, and *Loxosceles gaucho*. *L. laeta* is often considered the most dangerous of the recluse spiders, in part because it is the species that attains the largest body size. Recluses are found mostly inside homes, in basements, in attics, behind bookshelves and dressers, and in cupboards.⁹⁶

Loxosceles spiders bite and inoculate only a few microliters of venom to paralyze and digest their prey. “Loxoscelism” is a term used to describe lesions and reactions induced by bites from spiders of the genus *Loxosceles*. It causes skin necrosis, rhabdomyolysis, hemolysis, coagulopathy, AKI, and systemic inflammatory response syndrome (SIRS).⁹⁷

Loxoscelism is currently considered a serious public health problem in southern Brazil. There are 3000 reports of *Loxosceles* bites annually, and they constitute the third leading cause of accidents involving venomous animals.⁹⁸

The venom has a direct hemolytic effect on red blood cells and damages the membranes of endothelial cells in blood vessel walls.⁹⁹ Rhabdomyolysis and hemolysis are recognized cofactors involved in the genesis of AKI, and there is also evidence of a direct nephrotoxic action.¹⁰⁰ The clinical

manifestations include AKI, azotemia, oliguria, proteinuria, hematuria, and hemoglobinuria, all of them mainly related to rhabdomyolysis due to the destruction of muscle tissue. It is estimated that there are 90 reported cases of AKI related to loxocelism.¹⁰¹

Kidney biopsy specimens have shown hyalinization of tubular cells, erythrocytes in the Bowman space, glomerular collapse, interstitial edema, deposition of eosinophilic material in the tubule lumen, glomerular epithelial, and endothelial cell cytotoxicity and C3 in the mesangium.¹⁰² The biological activity of phospholipase D from *Loxosceles* venom may cause kidney lesions and AKI.¹⁰³ The major cause of death from loxocelism is attributed to AKI and its sequelae.

Treatment involves local wound care, pain management, and tetanus prophylaxis. Dapsone may be administered in some cases. Antivenoms for the treatment of recluse spider bites are available in Brazil, Mexico, and Peru.¹⁰⁴ Antivenom decreases the lesion size and limits systemic illness, even when its administration is delayed.¹⁰⁴ Immunization with inactive recombinant isoforms of phospholipase D may be a future option in the prevention of AKI associated with loxocelism.¹⁰⁴

Snake Envenomation

More than 2000 snake species exist worldwide, approximately 450 of which are venomous. It is estimated that out of the 2.4 million people bitten by venomous snakes each year, around 100,000 result in fatalities.¹⁰⁵ AKI is a significant contributor to these fatalities. An additional 400,000 individuals experience severe nonfatal health consequences, such as amputation, infection, tetanus, scarring, and contractures.¹⁰⁶

Approximately 12% of all venomous snakebites are complicated by AKI, with most cases occurring after bites by colubrids, elapids, or hydrophids.¹⁰⁷ The clinical manifestations depend on the snake species and amount of venom injected. Viper bites are characterized by pronounced local swelling, blistering, pain, tender regional lymphadenopathy, hemolysis, bleeding, and hypotension.¹⁰⁸ Sea snake venom is myotoxic, leading to myalgia and muscle weakness, and oliguria may develop. The incidence of snakebites is higher during rainy seasons. The age group most affected is individuals between 25 and 49 years old, with lower limbs being commonly injured sites.¹⁰⁸

Epidemiologic data on snakebite envenomation in LA are limited. In Brazil alone, there are approximately 20,000 venomous snakebite cases reported annually and a mortality rate of approximately 0.45%. *Bothrops* (pit viper) bites predominated, accounting for 96.6% of envenomations.¹⁰⁹ Finally, in Colombia, from 2008 to 2020 the population incidence of snake bites per 100,000 inhabitants was 9.5.¹¹⁰

Crotalus Snakes. AKI is frequently associated with *Crotalus* spp. in South America.^{111,112} The incidence of AKI ranges from 10% to 29% and typically occurs within the first 48 hours following the snakebite, attributed to rhabdomyolysis, kidney vasoconstriction venom exerts direct and indirect effects on kidney cells.¹¹² The histologic damage is acute tubular necrosis and interstitial nephritis. Contributing factors to

AKI may include shock, hypotension, hemolysis, and sepsis. In one study, dialysis was required in 24% of cases.¹¹² Mortality rates associated with AKI following *Crotalus* envenomation range from 8% to 17%.¹¹³

Bothrops Snakes. AKI associated with *Bothrops* envenomation has been linked to various factors including hemodynamic changes, myoglobinuria, hemoglobinuria, coagulation abnormalities, and direct nephrotoxicity of the venom.¹¹⁴

The venom can induce hemodynamic abnormalities through fluid sequestration at the bite site, bleeding, and the release of vasoactive substances.¹¹⁵ *Bothrops* venom has demonstrated hemolytic properties. Hemoglobinuria may contribute to AKI by exacerbating renal vasoconstriction, promoting glomerular coagulation, and inducing tubular nephrotoxicity.¹¹⁶

Kidney dysfunction manifests as oliguria, often necessitating dialysis in 33% to 75% of individuals. The most common kidney structural injury observed is acute tubular necrosis and, interstitial nephritis, and glomerulonephritis with mesangial proliferation have also been documented.¹¹⁷ The mortality rate associated with *Bothrops* venom-induced AKI ranges from 13% to 19%.¹¹⁸

Early administration of antivenom is crucial because it helps prevent or minimize the development of hematologic abnormalities, rhabdomyolysis, and AKI.¹¹⁹

Supportive measures encompass judicious volume replacement, administration of antitetanus immunoglobulin, and appropriate management of bleeding risks by utilizing fresh frozen plasma when necessary. Patients bitten by myotoxic snakes require aggressive fluid therapy and urinary alkalization to prevent intratubular precipitation of myoglobin.¹¹⁹ The estimated mortality rate from snakebite envenomation is up to 35%. A significant proportion of individuals, especially those with patchy or diffuse cortical necrosis and thrombotic microangiopathy, are left with varying degrees of kidney disfunction.

Lonomia Caterpillars

Caterpillars belonging to the *Lonomia* genus inhabit the trees within the lush forests of LA as their native environment.¹²⁰ These caterpillars possess a bristle-covered body, capable of injecting venom upon contact with a victim. Due to deforestation, these creatures have been pushed toward human settlements, resulting in a surge of accidents. The clinical manifestation is a localized burning sensation at the point of contact, headache, fever, vomiting, and even a potential bleeding diathesis, which could pose a significant risk to life.¹²¹ Instances of AKI linked to *Lonomia* venom have predominantly surfaced in Brazil. Administering antivenom early to rats has demonstrated the ability to avert hemorrhagic manifestations. The mechanisms underlying AKI encompass direct nephrotoxicity, intravascular hemolysis, activation of systemic and endothelial inflammatory pathways, hypotension, inflammation, heme-induced oxidative stress, coagulation, and activation of the complement system.¹²² Kidney biopsies have revealed acute tubular necrosis, deposition of fibrin within glomeruli, as well as instances of ischemia and tubular atrophy.¹²³ Notably, a cohort of five patients demonstrated the progression to CKD subsequent to AKI induced by *Lonomia* venom.¹²⁴

INDIGENOUS MEDICINE AND HERB USE AND ACUTE KIDNEY INJURY

The use of indigenous remedies of herbal or animal origin is an integral part of traditional cultures in many parts of LA. AKI resulting from ingestion of plant and fungal nephrotoxins is common and contributes to 18% of CA-AKI. Various kidney lesions include acute tubular necrosis, acute cortical necrosis, and AIN.¹²⁵ In LA, most cases of AKI are related to cat's claw (*Uncaria tomentosa*), a woody vine found in Amazon beds. It is believed that the oxindole alkaloids from the rot bark are those that provide medical effects against inflammatory processes (i.e., rheumatism). Cases of AIN have been reported after the use of cat's claw, as an idiosyncratic allergic reaction.^{126,127}

In the Andes region, an herb called Yacon (*Smallanthus sonchifolius*), used to treat diabetes, has been associated with AKI after prolonged exposure.¹²⁸ *L. meyenii* (Maca), a traditional herbal medicine used by indigenous cultures in South America for its supposed fertility enhancing, aphrodisiac (Andean Viagra), and nutritional properties, has been reported to cause AKI.¹²⁹ Starfruit is a rich source of oxalic acid, a nephrotoxic substance, and caramboxin, a potent neurotoxin, which are higher in *A. bilimbi* than *A. carambola*.¹³⁰ Risk factors associated with starfruit toxicity include CKD, the volume of starfruit ingestion, dehydration, and fasting. In CKD patients, toxic symptoms were observed after consuming approximately half a starfruit or around 25 mL. Individuals with normal kidney function can tolerate 4–6 starfruits or 300 mL of starfruit juice before experiencing toxicity.¹³¹ Circulating oxalates form calcium oxalate crystals, which can cause renal tubular obstruction, apoptosis of renal epithelial cells, interstitial inflammation, fibrosis, and nephron loss, and they ultimately contribute to the development of CKD.¹³² Early initiation of KRT or hemoperfusion is recommended to facilitate the clearance of oxalate and caramboxin, the toxic components of starfruit. Starfruit-associated AKI generally has a good prognosis.¹³³

PEDIATRIC KIDNEY DISEASE IN LATIN AMERICA

Pediatric nephrology in LA dates to the 1950s, with the founding of the pediatric nephrology department by Dr. Gustavo Gordillo-Paniagua at the Hospital Infantil de Mexico, followed by Drs. Carlos Gianantonio in Argentina and Julio Toporovsky in Brazil.¹³⁴ Currently there are residency programs in pediatric nephrology in Argentina, Brazil, Chile, Colombia, Costa Rica, Guatemala, Mexico, Uruguay, and Venezuela. In 2016, it was estimated that there were 850 pediatric nephrologists in the region (1.4 pmp).¹

EPIDEMIOLOGY OF CHRONIC KIDNEY DISEASE IN CHILDREN

Information on the epidemiology of CKD in children is scarce. Few countries have dialysis and most information comes from global registries, but it is often incomplete, especially from low-income countries. Among 4753 pediatric kidney transplants performed between 2004 and

2017, the main causes of treated ESKD were congenital anomalies of the kidney and urinary tract (31%), chronic glomerulopathies (24%), and hereditary nephropathies (13%).¹³⁴ In 14% of the patients the etiology was unknown. The most frequent histopathologic lesion was focal segmental glomerulosclerosis.¹³⁴

ACUTE KIDNEY INJURY IN CHILDREN

One literature review reported the main causes of AKI in children to be HUS (64%), cardiac disease (18%), and critical illness (17%), with a mortality rate of 27%.¹³⁵ CA-AKI represented 77% of the cases. Complete and partial recovery of renal function was observed in 59% and 33% of the patients, respectively.¹³⁵ Argentina is the country with the highest global incidence of HUS.¹³⁶ This is attributed to the high ingestion of meat and its early introduction into the diet in children. Indeed, in a study carried out on 72 healthy children, anti-Shiga toxin 1 and 2 antibodies were detected in 67% and 8% of the children studied, consistent with endemic Shiga toxin (Stx)-producing *Escherichia coli* (STEC) HUS in Argentina.¹³⁶

ACCESS TO DIALYSIS AND KIDNEY TRANSPLANTATION IN CHILDREN

As in the adult population, there are marked disparities in the access to KRT in the region. Access to dialysis and kidney transplantation is limited in 10 (47%) of the countries, mainly those without social security.¹³⁷ In El Salvador, Honduras, Nicaragua, Mexico, and Guatemala, <50% of children have access to RRT.^{138,17} Information on treatment modality and clinical outcomes is scarce. In Brazil (75%),¹³⁷ Chile (74%),¹³⁸ and Argentina (54%),¹³⁹ hemodialysis is the most frequent modality, while in Mexico (94%)¹⁴⁰ and Guatemala (60%),¹⁴¹ PD is the treatment of choice. The majority (64.4%) of patients were treated with automated peritoneal dialysis (APD) and reported outcomes are good.¹⁴²

In the latest (2018) registry report, 70 transplant centers were located across 18 countries.¹⁴² Between 2004 and 2017, the annual number of transplants doubled, from 200 to 400 procedures, resulting in a total of 4753 transplants, with more than half being performed in Brazil.¹⁴³ The mean age at transplant was 11.5 ± 3.6 years, and 12% were in children younger than 5 years of age. Fifty-four percent were female. The proportion of deceased-donor transplants increased from 56% to 82% in the same period.

Reported graft survival at 1 and 5 years was 94% and 90%, respectively. Five-year survival was 86% and 77% for living-donor and deceased-donor kidney transplantation, respectively. Patient survival at 1 and 5 years was 95% and 90%, respectively.¹⁴⁴ There are serious inequalities and gaps in the care of pediatric kidney patients in LA. To address these disparities, a multidimensional strategic model has been proposed to improve collaboration and knowledge sharing across the region.¹⁴⁴

TRENDS AND THE FUTURE

Kidney diseases pose significant health challenges in LA. CKD's prevalence is on the rise, driven by the increasing

incidence of risk factors such as diabetes, hypertension, and obesity in the region. Additionally, AKI remains a serious concern, often stemming from infectious diseases, environmental factors, and limited access to health care. Both CKD and AKI are further exacerbated by socioeconomic disparities and inadequate resources for prevention, early detection, and treatment.

Addressing these kidney health issues requires a comprehensive approach involving governments, health care systems, and communities. Increased investment in health care infrastructure, education, and public awareness campaigns can aid in the prevention and early detection of kidney diseases. Additionally, efforts to improve access to renal replacement therapies and dialysis services are crucial for managing CKD and AKI effectively.

Furthermore, fostering collaboration among health care providers, researchers, and policymakers is essential to drive advancements in kidney disease management and promote best practices across the region. By uniting efforts, LA can strive toward reducing the burden of CKD and AKI, improving patient outcomes, and enhancing the overall quality of life for its population.

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